

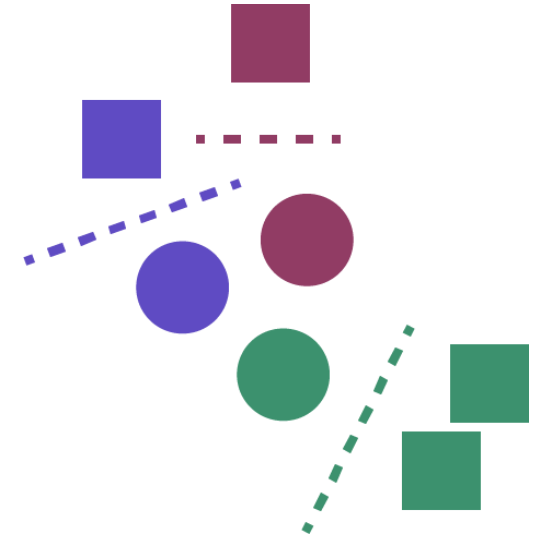
Group eXplanations with local approximation

Mattia Setzu *University of Pisa*

Riccardo Guidotti *University of Pisa*

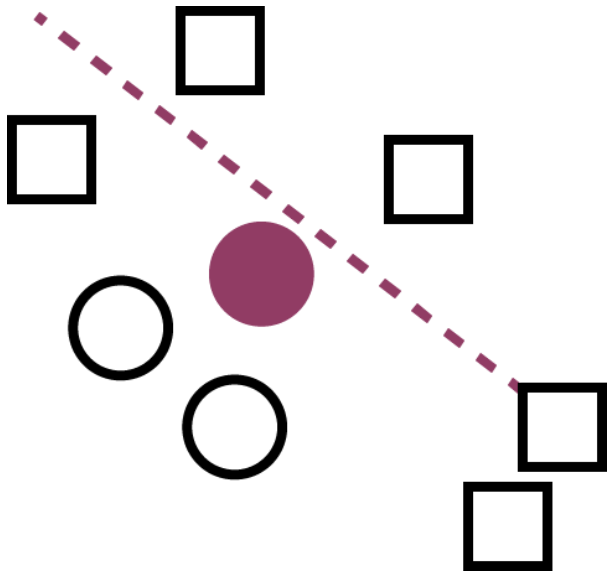
Dino Pedreschi *University of Pisa*

Fosca Giannotti *Scuola Normale Superiore*



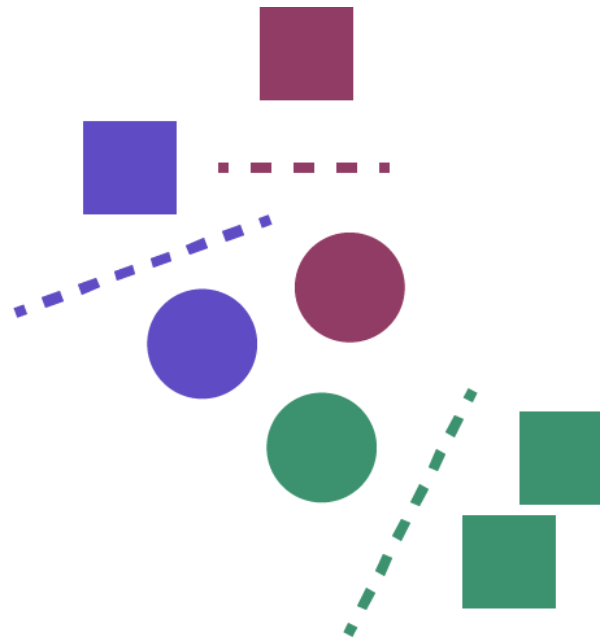
Explanation scope

Local



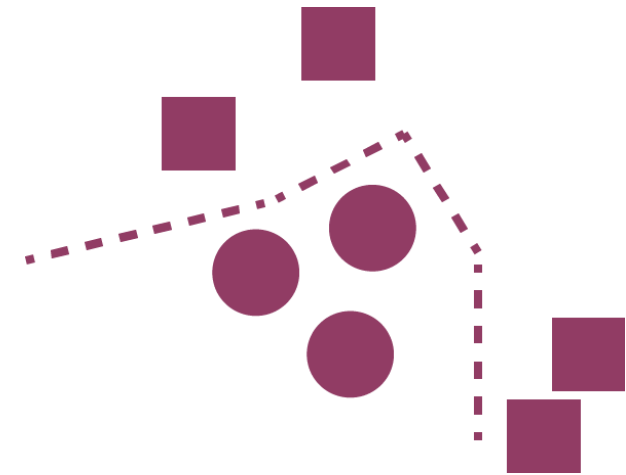
One instance, one
explaining model

Meso



One group, one
explaining model

Global



Many instances, one
explaining model

Meso-level: where are we?

Explain-first: bottom-up

First explain, then merge explanations

- Off the shelf local explanation algorithms

- Limited explanation type
- Computationally expensive

Group-first: top-down

Partition data into groups, then explain

- Cohesive groups
- Meso explanations by design

- Optimized for accuracy

Explainable AI for Trees: From Local Explanations to Global Understanding. Lundberg et al.

When Should We Use Linear Explanations? Delunay et al.

Interpretable Data Partitioning Through Tree-Based Clustering Methods. Guidotti et al.

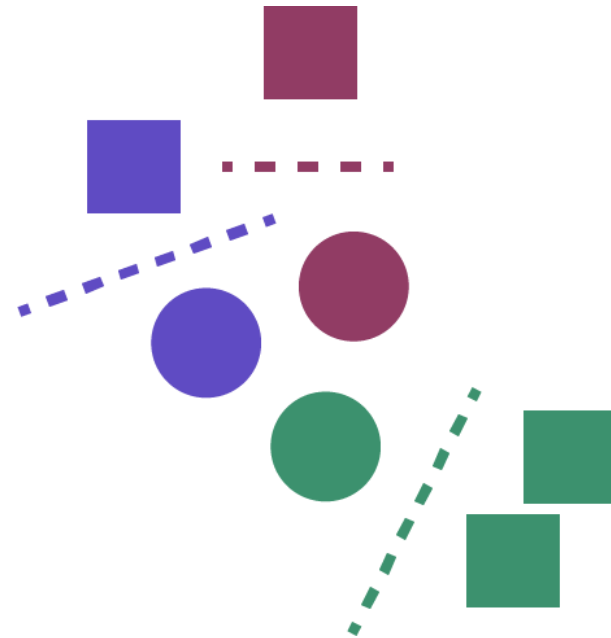
Faithful and Customizable Explanations of Black Box Models. Lakkaraju et al.

Meso-level explanations: GrouX

We propose a framework for meso explainability (**G**roup **eX**plainability).

It implements:

- *Feature importance* explanations by design
- Meso-level scope
- Flexible grouping



Meso-level: the two objectives

Discover meso groups

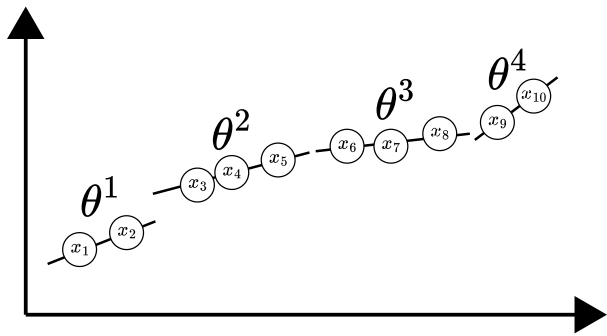
- Clustering discovers them w/ off-the shelf algorithms...
- ... that do not look for linear separability

Linear separability

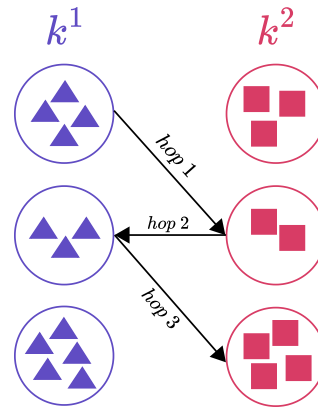
- We can test if instances in a cluster are linearly separable...
- ... but not if they are a good cluster

Key idea: disentangle the objective!

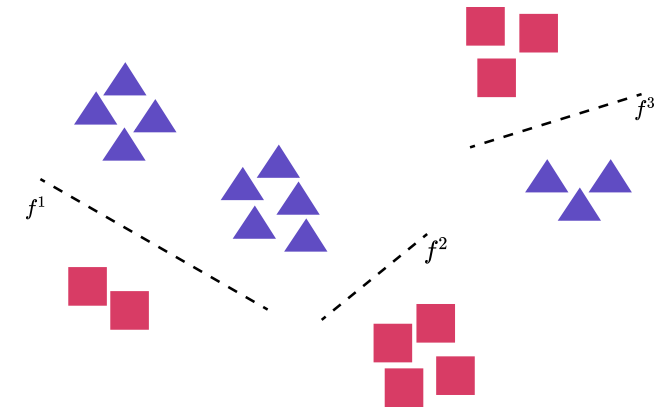
GrouX: the pipeline



1. Encode separability



2. Discover meso groups

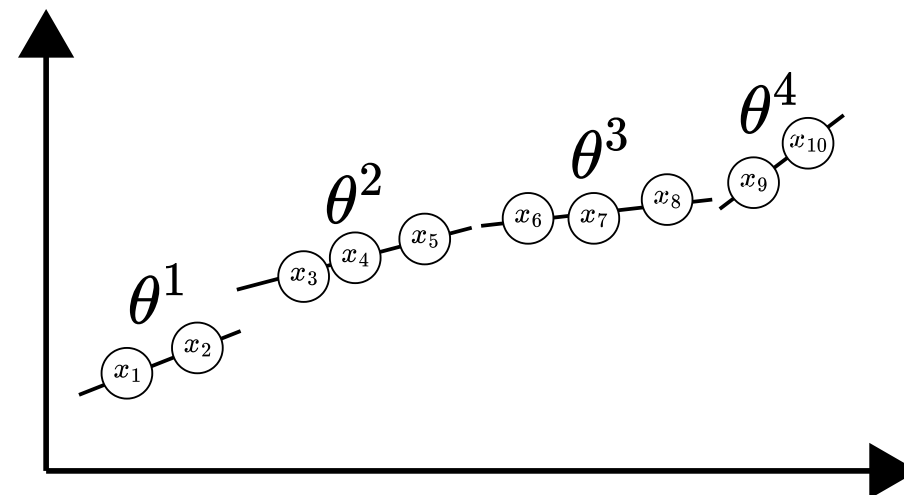


3. Learn meso models

1. Encode separability

How to encode linear separability?
With linear approximation

Instances with **similar values** are approximated by the same linear model, and the **same parameters**!

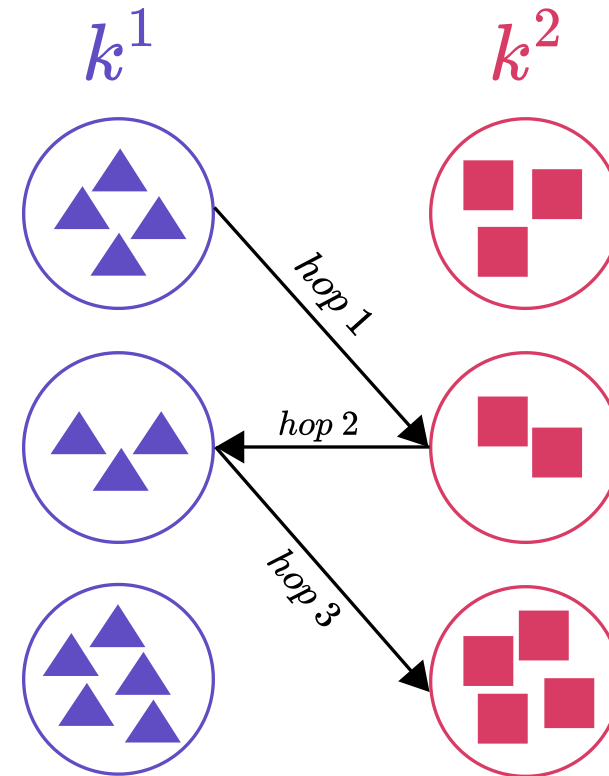


Piece-wise Linear Approximation (PLA): given a sequence of values, it approximates them with error-bounded linear splines.

2. Discover meso groups

We extract meso groups from the PLA representation through *Fair clustering* (Clusterlets).

Why *fair* clustering? Clustering is label-oblivious!

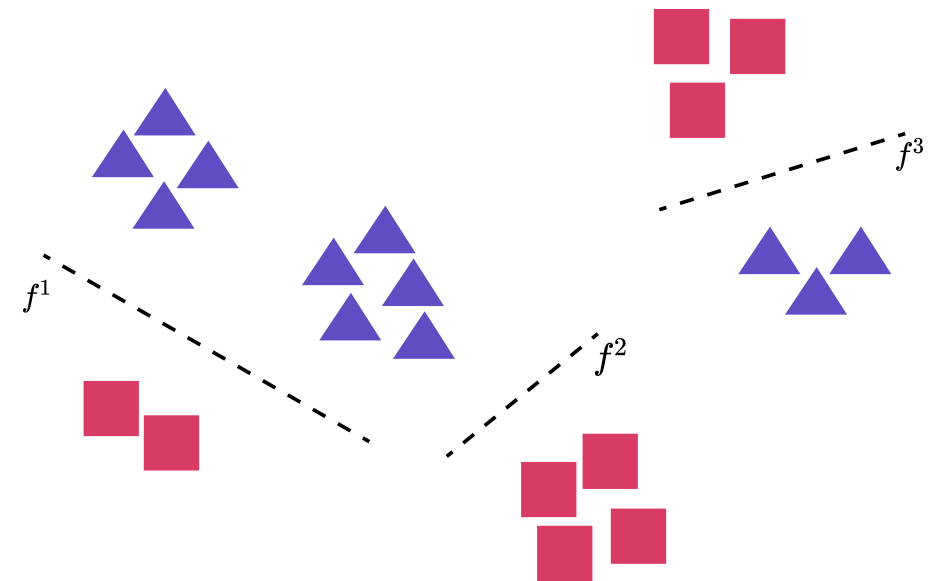


Creating fair clusters by merging mono-label clusterlets of classes k^1 (in blue) and k^2 (in red).

3. Learn meso-level models

Straightforward: learn a linear model, e.g., logistic regressor, on each meso model.

- **Explanation:** Linear model coefficients
- **Inference:** Closest meso model predicts



Meso-level linear models f^1, f^2, f^3 , learned on the respective meso-level group.

Experiments: meso models

How accurate and complex are the meso models?

Model selection per dataset, performance and number of meso models.

Model	F1	#meso models
<i>k</i> -means	.703 $\pm$.139	7.0 $\pm$ 6.1
ParTree	.693 $\pm$.160	5.0 $\pm$ 3.6
FParTree	.762 $\pm$.178	21.1 $\pm$ 54.7
GrouX	.841 $\pm$.178	22.4 $\pm$ 54.7

Experiments: Inference

Can meso models predict at global level?

GrouX VS Baseline VS Global interpretable models, model selection per dataset.

Model	F1	#meso models
<i>k</i> -means	.668 $\pm$.112	9.4 $\pm$ 4.9
ParTree	.663 $\pm$.086	4.2 $\pm$ 3.2
FParTree	.683 $\pm$.119	8.7 $\pm$ 6.4
GrouX	.738 $\pm$.112	16.2 $\pm$ 23.2
DT	.723 $\pm$.114	133 $\pm$ 149.0
LinearDT	.690 $\pm$.139	11.7 $\pm$ 3.71
Logistic	.676 $\pm$.135	

Inference... at smaller complexity

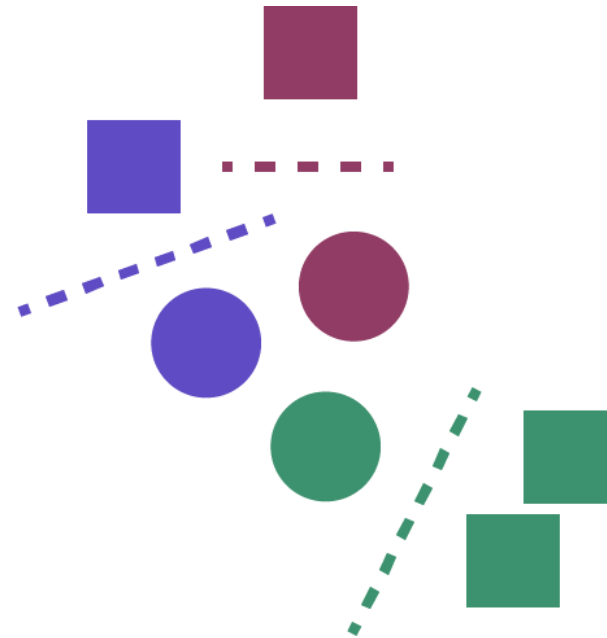
Can meso models predict at global level?

GrouX with different clustering k , model selection per dataset.

k	F1	#meso models
5	.810 $\pm$.148	5.6 $\pm$ 1.6
20	.867 $\pm$.131	22.8 $\pm$ 10.5
50	.895 $\pm$.124	57.8 $\pm$ 27.8

Next steps on group explainability

- **Model encoding** Can we learn representations for other explanation types, e.g., *decision rules*?
- **Discover meso groups** Can we provide descriptions of the meso groups?



GrouX: Group Explainability through Local Approximation

Interpretable by design model

- Meso level
- Linear explanations
- More accurate than competitors, and on par with global models
- Controllable complexity
- github.com/msetzu/groux

